

Lab Sheet 08

Title: Work with Aggregation Pipeline.

Objective:

- Understand and Apply MongoDB Aggregation Framework.
- Differentiate Between Key Aggregation Stages.
- Work with Accumulator Operators

Exercise 1:

Database: Garment

OrderID	CustomerName	Product	Category	Quantity	Price	OrderDate
G001	Anu	T-Shirt	Tops	3	20	2023-01-10
G002	Ravi	Jeans	Bottoms	1	40	2023-01-15
G003	Sara	Jacket	Outerwear	2	60	2023-01-20
G004	Anu	Skirt	Bottoms	1	25	2023-01-25
G005	Nina	Hoodie	Outerwear	1	55	2023-02-01
G006	Ravi	Shorts	Bottoms	2	30	2023-02-10
G007	Sara	Blouse	Tops	2	35	2023-02-14
G008	Nina	Coat	Outerwear	1	90	2023-02-20
G009	Anu	Shirt	Tops	2	25	2023-03-01
G010	Ravi	Skirt	Bottoms	1	25	2023-03-03
G011	Sara	Jacket	Outerwear	1	60	2023-03-05
G012	Nina	T-Shirt	Tops	2	20	2023-03-10

1. Insert all data in to the collection “garmentOrders”

```
// Use Database
use Garment

// 1. Insert all data into collection "garmentOrders"
db.garmentOrders.insertMany([
{
  OrderID: "G001",
  CustomerName: "Anu",
```

```
Product: "T-Shirt",
Category: "Tops",
Quantity: 3,
Price: 20,
OrderDate: new Date("2023-01-10")
},
{
  OrderID: "G002",
  CustomerName: "Ravi",
  Product: "Jeans",
  Category: "Bottoms",
  Quantity: 1,
  Price: 40,
  OrderDate: new Date("2023-01-15")
},
{
  OrderID: "G003",
  CustomerName: "Sara",
  Product: "Jacket",
  Category: "Outerwear",
  Quantity: 2,
  Price: 60,
  OrderDate: new Date("2023-01-20")
},
{
  OrderID: "G004",
  CustomerName: "Anu",
  Product: "Skirt",
  Category: "Bottoms",
  Quantity: 1,
  Price: 25,
  OrderDate: new Date("2023-01-25")
},
{
  OrderID: "G005",
  CustomerName: "Nina",
  Product: "Hoodie",
  Category: "Outerwear",
  Quantity: 1,
  Price: 55,
  OrderDate: new Date("2023-02-01")
},
{
  OrderID: "G006",
  CustomerName: "Ravi",
```

```
Product: "Shorts",
Category: "Bottoms",
Quantity: 2,
Price: 30,
OrderDate: new Date("2023-02-10")
},
{
  OrderID: "G007",
  CustomerName: "Sara",
  Product: "Blouse",
  Category: "Tops",
  Quantity: 2,
  Price: 35,
  OrderDate: new Date("2023-02-14")
},
{
  OrderID: "G008",
  CustomerName: "Nina",
  Product: "Coat",
  Category: "Outerwear",
  Quantity: 1,
  Price: 90,
  OrderDate: new Date("2023-02-20")
},
{
  OrderID: "G009",
  CustomerName: "Anu",
  Product: "Shirt",
  Category: "Tops",
  Quantity: 2,
  Price: 25,
  OrderDate: new Date("2023-03-01")
},
{
  OrderID: "G010",
  CustomerName: "Ravi",
  Product: "Skirt",
  Category: "Bottoms",
  Quantity: 1,
  Price: 25,
  OrderDate: new Date("2023-03-03")
},
{
  OrderID: "G011",
  CustomerName: "Sara",
```

```

    Product: "Jacket",
    Category: "Outerwear",
    Quantity: 1,
    Price: 60,
    OrderDate: new Date("2023-03-05")
  },
  {
    OrderID: "G012",
    CustomerName: "Nina",
    Product: "T-Shirt",
    Category: "Tops",
    Quantity: 2,
    Price: 20,
    OrderDate: new Date("2023-03-10")
  }
])

```

2.Show all orders in the "Tops" category.

```
db.garmentOrders.find({ Category: "Tops" })
```

3.Calculate total revenue from each category.

```

db.garmentOrders.aggregate([
{
  $group: {
    _id: "$Category",
    TotalRevenue: {
      $sum: { $multiply: ["$Quantity", "$Price"] }
    }
  }
}]

```

4.Find the highest quantity of any product ordered by a customer.

```

db.garmentOrders.aggregate([
{
  $group: {
    _id: null,
    HighestQuantity: { $max: "$Quantity" }
  }
}]

```

5. Calculate the total amount spent by each customer.

```
db.garmentOrders.aggregate([
{
  $group: {
    _id: "$CustomerName",
    TotalSpent: {
      $sum: { $multiply: ["$Quantity", "$Price"] }
    }
  }
}]
)
```

6. Show the average price of products in each category.

```
db.garmentOrders.aggregate([
{
  $group: {
    _id: "$Category",
    AveragePrice: { $avg: "$Price" }
  }
}]
)
```

7. Group by category and count how many products were sold.

```
db.garmentOrders.aggregate([
{
  $group: {
    _id: "$Category",
    TotalProductsSold: { $sum: "$Quantity" }
  }
}]
)
```

8. Sort customers by total amount spent in descending order.

```
db.garmentOrders.aggregate([
{
  $group: {
    _id: "$CustomerName",
    TotalSpent: {
      $sum: { $multiply: ["$Quantity", "$Price"] }
    }
  }
}]
)
```

```

},
{
  $sort: { TotalSpent: -1 }
}
])

```

9. Find all orders placed after February 1, 2023.

```

db.garmentOrders.find({
  OrderDate: { $gt: new Date("2023-02-01") }
})

```

10. Group products by category and find the maximum price.

```

db.garmentOrders.aggregate([
{
  $group: {
    _id: "$Category",
    MaxPrice: { $max: "$Price" }
  }
}
])

```

11. Save all "Outerwear" orders to a new collection outerwearOrders.

```

db.garmentOrders.aggregate([
{
  $match: { Category: "Outerwear" }
},
{
  $out: "outerwearOrders"
}
])

```

Exercise 02:

Database: Factory

SupplyID	SupplierName	Material	MaterialType	UnitsSupplied	UnitPrice	SupplyDate
S001	TexWorld	Cotton Roll	Fabric	100	5	2023-01-05
S002	SpinFast	Zippers	Accessories	200	0.5	2023-01-10
S003	TexWorld	Buttons	Accessories	500	0.2	2023-01-15
S004	ThreadX	Thread Pack	Thread	300	1	2023-01-20

S005	Colorify	Dye Pack	Dye	50	8	2023-01-25
S006	TexWorld	Cotton Roll	Fabric	150	5	2023-02-01
S007	SpinFast	Zipppers	Accessories	300	0.5	2023-02-05
S008	ThreadX	Thread Pack	Thread	400	1	2023-02-15
S009	TexWorld	Linen Roll	Fabric	100	6	2023-02-20
S010	Colorify	Dye Pack	Dye	60	8	2023-03-01
S011	ThreadX	Thread Pack	Thread	350	1	2023-03-05
S012	SpinFast	Buttons	Accessories	400	0.2	2023-03-10

1.Insert all data in to the collection “factorySupplies”

```
// Use Database
use Factory

// 1. Insert all data into collection "factorySupplies"
db.factorySupplies.insertMany([
{
  SupplyID: "S001",
  SupplierName: "TexWorld",
  Material: "Cotton Roll",
  MaterialType: "Fabric",
  UnitsSupplied: 100,
  UnitPrice: 5,
  SupplyDate: new Date("2023-01-05")
},
{
  SupplyID: "S002",
  SupplierName: "SpinFast",
  Material: "Zipppers",
  MaterialType: "Accessories",
  UnitsSupplied: 200,
  UnitPrice: 0.5,
  SupplyDate: new Date("2023-01-10")
},
{
  SupplyID: "S003",
  SupplierName: "TexWorld",
  Material: "Buttons",
  MaterialType: "Accessories",
  UnitsSupplied: 500,
  UnitPrice: 0.2,
```

```
    SupplyDate: new Date("2023-01-15")
  },
  {
    SupplyID: "S004",
    SupplierName: "ThreadX",
    Material: "Thread Pack",
    MaterialType: "Thread",
    UnitsSupplied: 300,
    UnitPrice: 1,
    SupplyDate: new Date("2023-01-20")
  },
  {
    SupplyID: "S005",
    SupplierName: "Colorify",
    Material: "Dye Pack",
    MaterialType: "Dye",
    UnitsSupplied: 50,
    UnitPrice: 8,
    SupplyDate: new Date("2023-01-25")
  },
  {
    SupplyID: "S006",
    SupplierName: "TexWorld",
    Material: "Cotton Roll",
    MaterialType: "Fabric",
    UnitsSupplied: 150,
    UnitPrice: 5,
    SupplyDate: new Date("2023-02-01")
  },
  {
    SupplyID: "S007",
    SupplierName: "SpinFast",
    Material: "Zippers",
    MaterialType: "Accessories",
    UnitsSupplied: 300,
    UnitPrice: 0.5,
    SupplyDate: new Date("2023-02-05")
  },
  {
    SupplyID: "S008",
    SupplierName: "ThreadX",
    Material: "Thread Pack",
    MaterialType: "Thread",
    UnitsSupplied: 400,
    UnitPrice: 1,
```



```

    SupplyDate: new Date("2023-02-15")
  },
  {
    SupplyID: "S009",
    SupplierName: "TexWorld",
    Material: "Linen Roll",
    MaterialType: "Fabric",
    UnitsSupplied: 100,
    UnitPrice: 6,
    SupplyDate: new Date("2023-02-20")
  },
  {
    SupplyID: "S010",
    SupplierName: "Colorify",
    Material: "Dye Pack",
    MaterialType: "Dye",
    UnitsSupplied: 60,
    UnitPrice: 8,
    SupplyDate: new Date("2023-03-01")
  },
  {
    SupplyID: "S011",
    SupplierName: "ThreadX",
    Material: "Thread Pack",
    MaterialType: "Thread",
    UnitsSupplied: 350,
    UnitPrice: 1,
    SupplyDate: new Date("2023-03-05")
  },
  {
    SupplyID: "S012",
    SupplierName: "SpinFast",
    Material: "Buttons",
    MaterialType: "Accessories",
    UnitsSupplied: 400,
    UnitPrice: 0.2,
    SupplyDate: new Date("2023-03-10")
  }
])

```

2. Show all supplies in the "Accessories" material type.

```

db.factorySupplies.find({
  MaterialType: "Accessories"
})

```

3. Calculate total cost (Units × Price) of supplies per material.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$Material",
      TotalCost: {
        $sum: {
          $multiply: [
            "$UnitsSupplied",
            "$UnitPrice"
          ]
        }
      }
    }
  }
])
```

4. Group by SupplierName and find the total number of units supplied.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$SupplierName",
      TotalUnits: { $sum: "$UnitsSupplied" }
    }
  }
])
```

5. Find the average unit price for each material type.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$MaterialType",
      AveragePrice: { $avg: "$UnitPrice" }
    }
  }
])
```

6. List the top 2 materials with the highest total cost.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$Material",
      TotalCost: {
        $sum: {
          $multiply: [
            "$UnitsSupplied",
            "$UnitPrice"
          ]
        }
      }
    },
    {
      $sort: { TotalCost: -1 }
    },
    {
      $limit: 2
    }
  ])
```

7.Count the number of different materials supplied by each supplier.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$SupplierName",
      Materials: { $addToSet: "$Material" }
    },
    {
      $project: {
        NumberOfMaterials: { $size: "$Materials" }
      }
    }
  ])
```

8. Show all supply records between Jan 20 and Feb 20, 2023.

```
db.factorySupplies.find({
  SupplyDate: {
    $gte: new Date("2023-01-20"),
    $lte: new Date("2023-02-20")
  }
})
```

9. Group by MaterialType and find the max and min unit prices.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$MaterialType",
      MaxPrice: { $max: "$UnitPrice" },
      MinPrice: { $min: "$UnitPrice" }
    }
  }
])
```

10. Calculate the total cost per supplier and sort by cost descending.

```
db.factorySupplies.aggregate([
  {
    $group: {
      _id: "$SupplierName",
      TotalCost: {
        $sum: {
          $multiply: [
            "$UnitsSupplied",
            "$UnitPrice"
          ]
        }
      }
    }
  },
  {
    $sort: { TotalCost: -1 }
  }
])
```

11. Output all “Fabric” supplies into a new collection fabricSupplies.

```
db.factorySupplies.aggregate([
  {
    $match: {
      MaterialType: "Fabric"
    }
  },
  {
    $out: "fabricSupplies"
  }
])
```